

Compute the quantities in Exercises 1-8 using the vectors

$$\mathbf{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$$

1. $\mathbf{u} \cdot \mathbf{u}$, $\mathbf{v} \cdot \mathbf{u}$, and $\frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$
2. $\mathbf{w} \cdot \mathbf{w}$, $\mathbf{x} \cdot \mathbf{w}$, and $\frac{\mathbf{x} \cdot \mathbf{w}}{\mathbf{w} \cdot \mathbf{w}}$
3. $\frac{1}{\mathbf{w} \cdot \mathbf{w}}$
4. $\frac{1}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$
5. $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \right) \mathbf{v}$ *scalar* $\rightarrow$ *vector*
6. $\left(\frac{\mathbf{x} \cdot \mathbf{w}}{\mathbf{x} \cdot \mathbf{x}} \right) \mathbf{x}$
7. $\|\mathbf{w}\|$
8. $\|\mathbf{x}\|$

$$1) \mathbf{u} \cdot \mathbf{u} = u_1 \cdot u_1 + u_2 \cdot u_2 = (-1) \cdot (-1) + 2 \cdot 2 = 5$$

$$\mathbf{v} \cdot \mathbf{u} = v_1 u_1 + v_2 u_2 = (-1) \cdot 4 + 2 \cdot 6 = 8$$

$$\frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \text{ (cannot cancel } \mathbf{u}) = \frac{8}{5} = 1.6$$

$$3) \frac{1}{\mathbf{w} \cdot \mathbf{w}} \cdot \mathbf{w} = \frac{1}{3^2 + 1^2 + 5^2} \cdot \begin{pmatrix} 3 \\ -1 \\ -5 \end{pmatrix} = \begin{bmatrix} 3/35 \\ -1/35 \\ -1/7 \end{bmatrix}$$

$$5) \frac{8}{52} \cdot \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 8/13 \\ 12/13 \end{bmatrix}$$

$$7) \|\mathbf{w}\| = \sqrt{\mathbf{w} \cdot \mathbf{w}} = \sqrt{35}$$

In Exercises 9-12, find a unit vector in the direction of the given vector.

$$9. \begin{bmatrix} -30 \\ 40 \end{bmatrix}$$

$$10. \begin{bmatrix} -6 \\ 4 \\ -3 \end{bmatrix} = \bar{\mathbf{a}}$$

$$11. \begin{bmatrix} 7/4 \\ 1/2 \\ 1 \end{bmatrix}$$

$$12. \begin{bmatrix} 8/3 \\ 2 \end{bmatrix} = \bar{\mathbf{b}}$$

$$13. \text{ Find the distance between } \mathbf{x} = \begin{bmatrix} 10 \\ -3 \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} -1 \\ -5 \end{bmatrix}.$$

$$10) \text{ if } \bar{\mathbf{a}} \text{ is a vector then } \frac{\bar{\mathbf{a}}}{\|\bar{\mathbf{a}}\|} \text{ is a unit vector}$$

$$\|\bar{\mathbf{a}}\| = \sqrt{36 + 16 + 9} = \sqrt{61}$$

$$\frac{\bar{\mathbf{a}}}{\|\bar{\mathbf{a}}\|} = \left[\frac{-6}{\sqrt{61}}; \frac{4}{\sqrt{61}}; \frac{-3}{\sqrt{61}} \right]^T$$

from Chapter 2:

22. Let $\mathbf{u} = (u_1, u_2, u_3)$. Explain why $\mathbf{u} \cdot \mathbf{u} \geq 0$. When is $\mathbf{u} \cdot \mathbf{u} = 0$?

$$12) \bar{\mathbf{b}} = \begin{bmatrix} 8/3 \\ 2 \end{bmatrix}, \quad \|\bar{\mathbf{b}}\| = \sqrt{\frac{64}{9} + 4} = \frac{\sqrt{64 + 36}}{3} = \frac{10}{3}$$

$$\frac{\bar{\mathbf{b}}}{\|\bar{\mathbf{b}}\|} = \left[\frac{8}{3} \cdot \frac{3}{10}; 2 \cdot \frac{3}{10} \right]^T = \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix}$$

$$\left\| \frac{\bar{\mathbf{b}}}{\|\bar{\mathbf{b}}\|} \right\| = \sqrt{\frac{16}{25} + \frac{9}{25}} = 1 \text{ - unit vector}$$

$$22) \quad \bar{\mathbf{u}} = (u_1, u_2, u_3) \quad a) \text{ Why } \bar{\mathbf{u}} \cdot \bar{\mathbf{u}} \geq 0?$$

$$b) \text{ When } \bar{\mathbf{u}} \cdot \bar{\mathbf{u}} = 0?$$

$$a) \quad \bar{\mathbf{u}} \cdot \bar{\mathbf{u}} = u_1^2 + u_2^2 + u_3^2 \text{ (sum of squares } \geq 0)$$

$$b) \quad u_1^2 + u_2^2 + u_3^2 = 0 \text{ if } u_1 = 0, u_2 = 0, u_3 = 0$$

23. Let $\mathbf{u} = \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -7 \\ -4 \\ 6 \end{bmatrix}$. Compute and compare $\mathbf{u} \cdot \mathbf{v}$, $\|\mathbf{u}\|^2$, $\|\mathbf{v}\|^2$, and $\|\mathbf{u} + \mathbf{v}\|^2$. Do not use the Pythagorean Theorem.

24. Verify the *parallelogram law* for vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$:

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

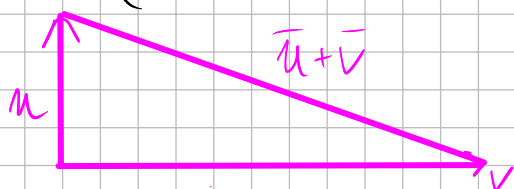
23) $\mathbf{u} \cdot \mathbf{v} = 2 \cdot (-7) - 5 \cdot (-4) - 1 \cdot 6 = -14 + 20 - 6 = 0$

$\mathbf{u}$ and $\mathbf{v}$ are orthogonal ($\perp$)

$$\|\bar{\mathbf{u}}\|^2 = \sqrt{\mathbf{u} \cdot \mathbf{u}}^2 = \mathbf{u} \cdot \mathbf{u} = 2^2 + 5^2 + 1^2 = 30$$

$$\|\bar{\mathbf{v}}\|^2 = \mathbf{v} \cdot \mathbf{v} = 7^2 + 4^2 + 6^2 = 49 + 16 + 36 = 101$$

$$\|\bar{\mathbf{u}} + \bar{\mathbf{v}}\|^2 = (\bar{\mathbf{u}} + \bar{\mathbf{v}}) \cdot (\bar{\mathbf{u}} + \bar{\mathbf{v}}) = \|\mathbf{u}\|^2 + 2 \cdot \bar{\mathbf{u}} \cdot \bar{\mathbf{v}} + \|\mathbf{v}\|^2 = 30 + 101 + 2 \cdot 0 = 131$$



24) $\|\bar{\mathbf{u}} + \bar{\mathbf{v}}\|^2 + \|\bar{\mathbf{u}} - \bar{\mathbf{v}}\|^2 = 2\|\bar{\mathbf{u}}\|^2 + 2\|\bar{\mathbf{v}}\|^2$

$$(\bar{\mathbf{u}} + \bar{\mathbf{v}})(\bar{\mathbf{u}} + \bar{\mathbf{v}}) + (\bar{\mathbf{u}} - \bar{\mathbf{v}})(\bar{\mathbf{u}} - \bar{\mathbf{v}}) \rightarrow 2 \cdot \|\bar{\mathbf{u}}\|^2 + 2\|\bar{\mathbf{v}}\|^2$$

$$\|\bar{\mathbf{u}}\|^2 + 2 \cdot \bar{\mathbf{u}} \cdot \bar{\mathbf{v}} + \|\bar{\mathbf{v}}\|^2 + \|\bar{\mathbf{u}}\|^2 - 2 \cdot \bar{\mathbf{u}} \cdot \bar{\mathbf{v}} + \|\bar{\mathbf{v}}\|^2 = 2\|\bar{\mathbf{u}}\|^2 + 2\|\bar{\mathbf{v}}\|^2$$

$$0 = 0$$

23. Let $\mathbf{u} = \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -7 \\ -4 \\ 6 \end{bmatrix}$. Compute and compare $\mathbf{u} \cdot \mathbf{v}$, $\|\mathbf{u}\|^2$, $\|\mathbf{v}\|^2$, and $\|\mathbf{u} + \mathbf{v}\|^2$. Do not use the Pythagorean Theorem.

24. Verify the *parallelogram law* for vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$:

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

23) $\mathbf{u} \cdot \mathbf{v} = 0$, $\|\mathbf{u}\|^2 = 30$, $\|\mathbf{v}\|^2 = 101$

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \left\| \begin{bmatrix} 2-7 \\ -5-4 \\ -1+6 \end{bmatrix} \right\|^2 = \left\| \begin{bmatrix} -5 \\ -9 \\ 5 \end{bmatrix} \right\|^2 = \\ &= 25 + 81 + 25 = 131 \end{aligned}$$

24) $\mathbf{u} = (u_1, u_2, u_3, \dots, u_n)$ $\mathbf{v} = (v_1, \dots, v_n)$

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \left\| (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n) \right\|^2 = \\ &= (u_1 + v_1)^2 + (u_2 + v_2)^2 + \dots + (u_n + v_n)^2 \end{aligned}$$

$$\|\mathbf{u} - \mathbf{v}\|^2 = (u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2$$

$$\|\mathbf{u}\|^2 = u_1^2 + u_2^2 + \dots + u_n^2$$

$$\|\mathbf{v}\|^2 = v_1^2 + v_2^2 + \dots + v_n^2$$

$$\begin{aligned} & \overset{2u_1v_1}{(u_1+v_1)^2} + \dots + (u_n+v_n)^2 + \overset{-2u_1v_1}{(u_1-v_1)^2} + \dots + (u_n-v_n)^2 = 2(u_1^2 + \dots + u_n^2 + v_1^2 + \dots + v_n^2) \\ & \Rightarrow 0 = 0 \end{aligned}$$

25. Let $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$. Describe the set H of vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ that are orthogonal to $\mathbf{v}$. [Hint: Consider $\mathbf{v} = \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$.]

$$H = \{ \mathbf{h} \mid \mathbf{h} = \begin{bmatrix} x \\ y \end{bmatrix} \}$$

26. Let $\mathbf{u} = \begin{bmatrix} 5 \\ -6 \\ 7 \end{bmatrix}$, and let W be the set of all $\mathbf{x}$ in $\mathbb{R}^3$ such that $\mathbf{u} \cdot \mathbf{x} = 0$. What theorem in Chapter 4 can be used to show that W is a subspace of $\mathbb{R}^3$? Describe W in geometric language.

25) $\mathbf{v} = \mathbf{0} \Leftrightarrow a = 0, b = 0$
 $\mathbf{v} \cdot \mathbf{h} = 0$ - are orthogonal

(A) $a \cdot x + b \cdot y = 0$ if $a = 0, b \neq 0 \Rightarrow x, y$ - any numbers
 $H = \mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \mid x \in \mathbb{R}, y \in \mathbb{R} \right\}$

(B) $ax + by = 0$ if $a = 0, b \neq 0$
 $b \cdot y = 0 \rightarrow y = 0$

$H_x = H = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} \mid x \in \mathbb{R} \right\}$ set of vectors on Ox

(C) $ax + by = 0$ if $a \neq 0, b = 0 \Rightarrow x = 0$

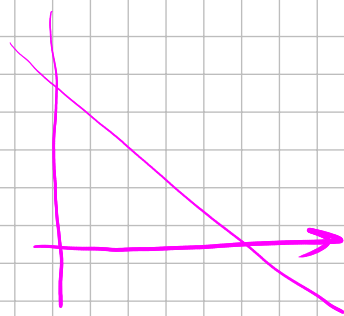
$H = \left\{ \begin{bmatrix} 0 \\ y \end{bmatrix} \mid y \in \mathbb{R} \right\}$ - Oy

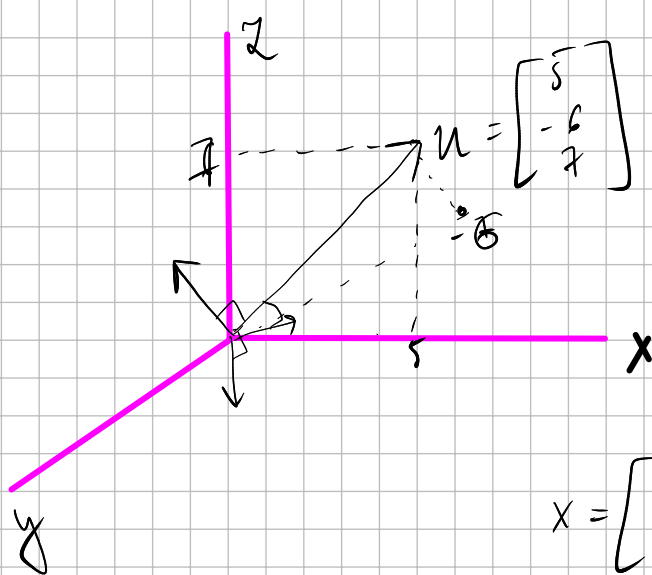
(D) $ax + by = 0$ if $a \neq 0, b \neq 0$

$$x = -\frac{b}{a}y$$

$H = \left\{ \begin{bmatrix} -\frac{b}{a}y \\ y \end{bmatrix} \mid y \in \mathbb{R} \right\}$ - Line

passing through the origin





$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$X : X \cdot u = 0$$



$$5 \cdot x_1 - 6 \cdot x_2 + 7 \cdot x_3 = 0$$

$$5x - 6y + 7z = 0$$

equation of a plane
in $\mathbb{R}^3$

32. (M) Construct a pair $\underline{u}, \underline{v}$ of random vectors in $\mathbb{R}^4$, and let

$$\underline{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\underline{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} .5 & .5 & .5 & .5 \\ .5 & .5 & -.5 & -.5 \\ .5 & -.5 & .5 & -.5 \\ .5 & -.5 & -.5 & .5 \end{bmatrix} \quad \begin{array}{l} \|a_1\| = \sqrt{(\frac{1}{2})^2 \cdot 4} = 1 \\ \|a_1\| = 1, \|a_2\| = 1, \|a_3\| = 1 \\ a_1 \cdot a_2 = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} = 0 \\ a_3 \cdot a_4 = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} = 0 \end{array} \left. \begin{array}{l} \text{all are} \\ \text{unit vectors} \end{array} \right\} \begin{array}{l} \text{all} \\ \text{are} \\ \text{orthogonal} \end{array}$$

- a. Denote the columns of A by $\underline{a}_1, \dots, \underline{a}_4$. Compute the length of each column, and compute $\underline{a}_1 \cdot \underline{a}_2$, $\underline{a}_1 \cdot \underline{a}_3$, $\underline{a}_1 \cdot \underline{a}_4$, $\underline{a}_2 \cdot \underline{a}_3$, $\underline{a}_2 \cdot \underline{a}_4$, and $\underline{a}_3 \cdot \underline{a}_4$.

A - orthogonal matrix

- b. Compute and compare the lengths of $\underline{u}$, $A\underline{u}$, $\underline{v}$, and $A\underline{v}$.

- c. Use equation (2) in this section to compute the cosine of the angle between $\underline{u}$ and $\underline{v}$. Compare this with the cosine of the angle between $A\underline{u}$ and $A\underline{v}$.

- d. Repeat parts (b) and (c) for two other pairs of random vectors. What do you conjecture about the effect of A on vectors?

$$\|A\underline{x}\| = \|\underline{x}\| \quad ; \quad \|A\underline{u}\| = \|\underline{u}\|$$

$$\begin{bmatrix} 0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & -0.5 & 0.5 & 0.5 \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\|\underline{u}\| = \sqrt{1^2 + (-1)^2 + 1^2 + (-1)^2} = 2$$

$$\|A \cdot \underline{u}\| = \sqrt{2^2 + 1^2 + 1^2 + 1^2} = 2$$